ModelIt!

Teacher's Guide

The Coral Reef Death Spiral

Modeling Thermal Bleaching, Recovery Thresholds, and Climate Feedback

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-LS2-2, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-2, HS-LS2-6

Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical relationships between bleaching frequency and recovery time to predict coral population trajectories and identify critical thresholds for reef survival.
Disciplinary Core Idea	**LS2.A Interdependent Relationships / LS2.C Ecosystem Dynamics** — Students model the mutualistic symbiosis between corals and zooxanthellae and analyze how disrupting this relationship cascades through the reef ecosystem to affect biodiversity at every trophic level.
Crosscutting Concept	**Stability and Change / Scale, Proportion, and Quantity** — Students analyze how small temperature changes (1-2 degrees C) at the global scale produce dramatic ecosystem transformations at the reef scale, and how the quantitative relationship between stress frequency and recovery time determines long-term stability.

CAST Testing Alignment

MULTIPLE CHOICE:

A coral reef experiences bleaching events in 2020, 2024, and 2027. Full recovery from a bleaching event requires 10-15 years. Based on your model, what is the most accurate prediction for this reef's trajectory?

CONSTRUCTED RESPONSE:

Using your model, explain why a reef that survived a severe bleaching event in 2016 might not survive a less severe event in 2022. Reference bleaching frequency, recovery time, Reef Structural Complexity, and Algae Competition in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Coral reefs are often called the 'rainforests of the sea' — they cover less than 1% of the ocean floor but support approximately 25% of all marine species. They also protect coastlines from storm damage, support fisheries feeding 500 million people, and generate $375 billion annually in economic value. Yet these ecosystems are now dying at an unprecedented rate, driven by a death spiral of increasing thermal stress and shrinking recovery windows.

The Coral-Zooxanthellae Symbiosis

Corals are animals — colonies of tiny polyps related to jellyfish — that host photosynthetic algae called zooxanthellae within their tissues. This mutualistic relationship is one of the most important in the ocean: zooxanthellae provide up to 90% of the coral's energy through photosynthesis, while corals provide shelter, CO2, and nutrients. The partnership is temperature-sensitive: when water temperatures exceed 1-2 degrees C above the local summer maximum for 4+ weeks, the algae produce toxic reactive oxygen species. Corals expel their zooxanthellae in self-defense — the process called bleaching — but lose their primary energy source in the process.

Bleaching Frequency and Recovery Dynamics

In the 1980s, mass bleaching events occurred approximately once every 25-30 years, giving reefs ample time to recover. By the 2010s, the average interval had shrunk to 6 years. The Great Barrier Reef bleached in 2016, 2017, 2020, 2022, 2024, and 2025 — essentially continuous stress with no recovery window. Full recovery from a severe bleaching event requires 10-15 years under optimal conditions. When bleaching occurs before recovery is complete, each event starts from a weaker baseline. The mathematics are stark: if recovery requires 10 years but bleaching occurs every 5 years, the reef loses ground with every cycle.

Phase Shifts and Irreversibility

When coral cover drops below approximately 10-20% of the reef surface, a critical tipping point is crossed. Dead coral skeletons are rapidly colonized by macroalgae, which grow faster than coral and prevent coral larvae from settling. Herbivorous fish that would normally control algae are often depleted by overfishing. The reef undergoes a 'phase shift' from a coral-dominated, high-biodiversity ecosystem to an algae-dominated, low-biodiversity system. This transition is functionally irreversible on human timescales because the conditions for coral recolonization no longer exist — it would require eliminating algae, restoring herbivores, and reducing temperatures simultaneously.

Global Consequences of Reef Loss

The stakes extend far beyond the reef itself. An estimated 500 million people depend on coral reef fisheries for protein and income. Reef structures provide coastal protection worth $9.9 billion per year by absorbing up to 97% of wave energy during storms — without reefs, coastal flooding and erosion increase dramatically. Reef ecosystems are estimated to contain over 1 million species, many undiscovered, with significant pharmaceutical potential. The loss of coral reefs would represent the first complete collapse of a global biome in human history.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the coral reef system: Sea Surface Temperature, Bleaching Frequency, Coral Recovery Rate, Reef Structural Complexity, Marine Biodiversity, and Algae Competition — distinguishing the external thermal driver from internal ecosystem responses.

E **Establish Relationships**

Students determine that rising Sea Surface Temperature increases Bleaching Frequency, which reduces Coral Recovery Rate by repeatedly damaging corals before they heal. Declining recovery degrades Reef Structural Complexity, which reduces Marine Biodiversity and creates space for Algae Competition — which further inhibits coral recovery in a reinforcing loop.

V **Visualize & Model**

Students build a computational model showing how the timing mismatch between bleaching frequency and recovery time creates a downward spiral that accelerates toward phase shift.

E **Evaluate Outcomes**

Students run historical, current, and accelerated warming scenarios to identify the bleaching frequency threshold at which recovery becomes impossible and phase shift is triggered.

R **Revise & Extend**

Students add Ocean Acidification, Herbivore Fish Population, or Coral Genetic Diversity to explore how additional stressors and adaptive capacity affect the reef's tipping point.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This reef survived ice ages, asteroid impacts, and 500 million years of evolution. But it might not survive the next 30 years. Here's why."*

Watch For

Show time-lapse of coral bleaching from vibrant to ghostly white. Ask: What do you think caused this transformation? How long do you think it took?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling a death spiral — a system that crosses a point of no return. And it's happening right now to the most biodiverse ecosystems in the ocean."*

Watch For

Have students read objectives. Pre-teach 'symbiosis' and 'phase shift.' Show a healthy coral close-up — point out the tiny zooxanthellae giving it color. Then show the same species bleached.

Slide 3: Big Question

3 min

Say: *"One to two degrees. That's all it takes. One to two degrees above normal for a few weeks, and the entire relationship between coral and algae — 240 million years old — falls apart."*

Watch For

Quick math exercise: If bleaching happened every 25 years in the 1980s and now happens every 6 years, how many times will today's reefs bleach by 2050? How many recovery cycles does that allow?

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of the most important symbiotic relationship in the ocean — and what happens when it breaks."*

Watch For

Preview each LEVER step. Emphasize that this system has a tipping point — a threshold beyond which the damage becomes self-reinforcing and irreversible.

Slide 5: Activity 1: Components

8 min

Say: *"One external driver — temperature. Five internal responses. But here's the key: some of those responses feed back into each other, creating a spiral."*

Watch For

Guide sorting of external vs. internal components. Discuss why only Sea Surface Temperature is external — everything else responds to the cascade. Preview the algae-coral competition dynamic.

Slide 6: Activity 2: Connections

8 min

Say: *"Find the death spiral. Temperature triggers bleaching. Bleaching prevents recovery. Failed recovery degrades the reef. Degraded reef gets overgrown by algae. Algae prevent coral from coming back. Where does it end?"*

Watch For

Help students map the reinforcing feedback loop. The critical insight is that Algae Competition feeds back to inhibit Coral Recovery Rate, creating a self-reinforcing spiral. Identify the tipping point.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's rewind to the 1980s when reefs were thriving. Then let's fast-forward through the damage and see where the death spiral begins."*

Watch For

Students predict the bleaching frequency threshold for phase shift before running simulations. Compare the three scenarios. Identify exactly when recovery rate can no longer match bleaching frequency.

Slide 8: Discoveries

5 min

Say: *"Your model just proved that it's not about single events — it's about the ratio of damage to healing. And right now, that ratio is fatal for most reefs."*

Watch For

Lead discussion on irreversibility and tipping points. Connect to the Great Barrier Reef's back-to-back bleaching events. Discuss what 500 million people lose if reefs collapse.

Slide 9: STEM Challenge

12 min

Say: *"You can't turn down the ocean's thermostat. So what CAN you do to buy these reefs time? Design a strategy that gives corals their best shot at surviving the next 30 years."*

Watch For

Groups design comprehensive resilience plans addressing local stressors, coral nurseries, marine protected areas, and heat-tolerant coral propagation. Present as conservation agency proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sea Surface Temperature
- **INTERNAL:** Bleaching Frequency, Coral Recovery Rate, Reef Structural Complexity, Marine Biodiversity, Algae Competition

Relationships

- **Sea Surface Temperature to Bleaching Frequency:** POSITIVE (+) — Higher Sea Surface Temperature causes more frequent exceedances of the coral thermal stress threshold, directly increasing Bleaching Frequency. Each 0.5 degree C increase in average temperature roughly doubles the frequency of thermal stress events.
- **Bleaching Frequency to Coral Recovery Rate:** NEGATIVE (-) — More frequent bleaching events reduce Coral Recovery Rate because each event damages corals before they have recovered from the previous one. With insufficient recovery time, corals enter each new stress event in a weakened state.
- **Coral Recovery Rate to Reef Structural Complexity:** POSITIVE (+) — When corals recover successfully, they rebuild three-dimensional reef structure through skeletal growth. When recovery rate drops below the rate of erosion and bioerosion, structural complexity declines as dead coral crumbles.
- **Reef Structural Complexity to Marine Biodiversity:** POSITIVE (+) — Complex reef architecture provides the habitat niches — caves, crevices, overhangs, and branching structures — that support diverse communities. As structure simplifies, species requiring specific habitats are lost, reducing biodiversity.
- **Algae Competition to Coral Recovery Rate:** NEGATIVE (-) — Macroalgae overgrowth on dead coral surfaces physically prevents coral larvae from settling and growing, creating a feedback loop that further reduces Coral Recovery Rate and locks the reef in a degraded state.

Key Discoveries

- The coral reef death spiral is driven by a timing mismatch: bleaching events are becoming more frequent (every 6 years) while recovery requires 10-15 years — each event compounds damage from the previous one
- Phase shifts from coral to algae dominance are often irreversible on human timescales because algae prevent coral larvae from settling and recolonizing — the reef gets locked into a degraded state
- The Great Barrier Reef has experienced five mass bleaching events since 2016 — a frequency that gives corals virtually no recovery time between events
- Reef collapse eliminates not just corals but the entire three-dimensional habitat structure that supports 25% of marine species, eliminates coastal storm protection, and removes a significant carbon sink

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bleached coral is dead coral"	Bleaching is a stress response, not death — bleached corals have expelled their zooxanthellae but are still alive. If temperatures return to normal within a few weeks, corals can reacquire zooxanthellae and recover. Death occurs when bleaching persists too long and corals starve without their photosynthetic partners.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Coral reefs only matter for tourism"	Coral reefs provide far more than scenic diving destinations. They protect coastlines from storms (absorbing 97% of wave energy), support fisheries feeding 500 million people, contain compounds used in cancer and HIV treatments, and cycle nutrients that support the broader ocean ecosystem. Their economic value is estimated at $375 billion per year — tourism is only a fraction of that.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can just build artificial reefs to replace natural ones"	Artificial reefs (sunken ships, concrete structures) provide hard substrate but lack the biological complexity of living coral reefs. They don't photosynthesize, don't build three-dimensional architecture that grows and adapts, don't support the same biodiversity, and don't provide coastal protection at scale. Artificial reefs can supplement conservation efforts but cannot replace the ecological function of living coral.